Credit Card Fraud Detection Using Machine Learning

PROJECT REPORT

Submitted by

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Submitted to: _____

Institution / Department: _____

Academic Year: 2026

Abstract

Credit card fraud can cause financial loss and reduce customer trust. This project applies supervised machine learning to classify potentially fraudulent transactions in an imbalanced dataset. SMOTE was used to address class imbalance on training data, and Logistic Regression, Random Forest, and XGBoost were compared. Random Forest + SMOTE achieved the highest F1-score (0.8629) and precision (0.8586), while XGBoost achieved the highest ROC-AUC (0.9785) and PR-AUC (0.8731).

1. Introduction

Fraud detection aims to identify suspicious transactions while limiting false alarms. Since fraud is rare, accuracy alone is not sufficient; precision, recall, F1-score, ROC-AUC, and PR-AUC provide a more informative evaluation.

2. Problem Statement

Build a classification model that distinguishes fraudulent from legitimate credit card transactions, addresses class imbalance, compares suitable algorithms, and selects a model based on measured performance.

3. Objectives

• Explore dataset quality and class imbalance.
• Inspect duplicate rows and transaction features.
• Apply SMOTE only to training data.
• Compare Logistic Regression, Random Forest, and XGBoost.
• Evaluate results with precision, recall, F1-score, ROC-AUC, and PR-AUC.

4. Dataset

The original Kaggle dataset contains 284,807 transactions and 31 columns, including 492 fraud cases. A data-quality check identified 1,081 duplicate rows. The latest reported test split contains 56,962 transactions, including 98 fraud cases. Confirm the final notebook's preprocessing and split before claiming the metrics were computed after duplicate removal. The target is Class (0 = legitimate, 1 = fraud); features include Time, Amount, and anonymized V1–V28 variables.

5. Methodology

The workflow includes data inspection, train-test splitting, handling class imbalance with SMOTE, training three classifiers, and evaluating on a held-out test set. SMOTE must be applied only to training data to prevent leakage. Logistic Regression is a baseline, Random Forest combines decision trees, and XGBoost uses gradient-boosted trees.

6. Model Comparison

The following values are the latest recorded results. Verify them against the final executed notebook before submission. Random Forest + SMOTE had the highest F1-score and precision. XGBoost had the highest ROC-AUC and PR-AUC, so the preferred model depends on the cost of missed fraud versus false alerts.

7. Confusion Matrix

The recorded Random Forest confusion matrix contains 56,850 true negatives, 14 false positives, 13 false negatives, and 85 true positives. Thus, 85 of 98 fraud cases were detected and 13 were missed.

8. Conclusion

The project demonstrates data exploration, imbalance handling, model training, and metric-based comparison. Random Forest + SMOTE led on F1-score and precision, while XGBoost led on ROC-AUC and PR-AUC. Multiple metrics are important when evaluating fraud detection.

9. Limitations and Future Work

Anonymized features limit interpretation. Fraud patterns may change over time. Future work could tune decision thresholds, evaluate business costs, monitor model drift, and retrain on newer data. Ensure the final notebook's metrics match the

reported results.

References

1. Kaggle Credit Card Fraud Detection: https://www.kaggle.com/datasets/mlg-ulb/creditcardfraud
2. Chawla et al. (2002), SMOTE: Synthetic Minority Over-sampling Technique, Journal of Artificial Intelligence Research.
3. scikit-learn documentation: https://scikit-learn.org/
4. XGBoost documentation: https://xgboost.readthedocs.io/

Model Comparison Metrics

Model	Precision	Recall	F1-score	ROC-AUC	PR-AUC
Logistic Regression + SMOTE	0.5181	0.8776	0.6515	0.9740	0.7183
Random Forest + SMOTE	0.8586	0.8673	0.8629	0.9643	0.8656
XGBoost + SMOTE	0.7265	0.8673	0.7907	0.9785	0.8731

Random Forest Confusion Matrix

	Predicted Legitimate	Predicted Fraud
Actual Legitimate	56,850 (TN)	14 (FP)
Actual Fraud	13 (FN)	85 (TP)